

Chapter 1

Functions and Relations

1.1 Functions

The previous chapter covered sets and their basic operations. Now we introduce the notion of a function between sets. Just as in calculus, a function will assign a unique output for every input. Familiar notations such as, f and $f(x)$ will be used. The same concepts will be used; only the inputs and outputs will include things other than numbers. A *function* (or *mapping*) from sets A to B , $f: A \rightarrow B$, is a rule that assigns each element in A to exactly one element in B .

- A is called the *domain* of f .
 - B is called the *range* of f .¹
 - If $f(a) = b$ for some $a \in A$, then b is called the *image of a under f* .
 - $\{b: b \in B \text{ and } f(a) = b \text{ for some } a \in A\}$ is called the *image of A under f* .
- Functions are well defined. Meaning that an input is assigned to one and only output.

Example 1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$. Then $f(x) = x^2 + 1$ is a function. it is clear from the equation that f is well defined.

Example 2. Let $f: \mathbb{R} \rightarrow \mathbb{R}$. Then $f(x) = \pm\sqrt{x}$ is not a function. Since $f(1) = +1$ and -1 .

Example 3. Consider $f: \mathbb{R} \rightarrow \mathbb{Z}$ given by the *floor function* $f(x) = \lfloor x \rfloor$ where $\lfloor x \rfloor$ is the largest integer less than or equal to x , e.g., $\lfloor \pi \rfloor = 3$. Note that the floor function is also a function from $\mathbb{R}$ to $\mathbb{R}$, $f: \mathbb{R} \rightarrow \mathbb{Z}$.

Example 4. Let $f: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(x, y) \rightarrow xy$. Then f is a function.

Example 5. Let $A = \{a, b, c, \dots, z\}$ and $f: A \rightarrow \mathbb{Z}$ be function defined by assigning the n^{th} letter to n , e.g., $f(3) = c$.

Definition 1. Let $f: B \rightarrow C$ and $g: A \rightarrow B$ be functions, then the *composition* of f and g is $f \circ g: A \rightarrow C$ defined as $f(g(a)) = c$ where $g(a) = b$ and $f(b) = c$ for elements $a \in A$, $b \in B$, and $c \in C$.

- In general the composition of function is not commutative, i.e., $f \circ g \neq g \circ f$.

Example 6. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ and $f(x) = 2x + 1$ and $g(x) = x^2$. Then $f \circ g(x) = f(g(x)) = 2(x^2) + 1 = 2x^2 + 1$.

Definition 2. A function, $f: A \rightarrow B$, is called *one-to-one*² if $f(a_1) = f(a_2)$ implies that $a_1 = a_2$.

¹Also called the *co-domain*.

²Also called an *injective function*

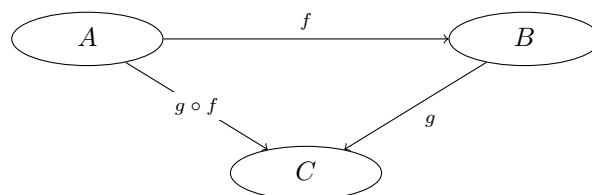


Figure 1.1: Composition of functions f and g .

Example 7. $f: \mathbb{Z} \rightarrow \mathbb{Z}$ where $f(x) = x^3$ is one-to-one.

Proof. Suppose that $f(x_1) = f(x_2)$ for some $x_1, x_2 \in \mathbb{Z}$. So $(x_1)^3 = (x_2)^3 \Rightarrow ((x_1)^3)^{1/3} = ((x_2)^3)^{1/3} \Rightarrow x_1 = x_2$ $\square$

Exercise 1. Show that any linear function $f: \mathbb{R} \rightarrow \mathbb{R}$ with a non-zero defined slope is one-to-one.

Example 8. $f: \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = |x|$ is not one-to-one.

$|-1| = 1 = |1|$ but $-1 \neq 1$ so f is not one-to-one.

Proposition 1. If $f: A \rightarrow B$ then $f^{-1} = \{x: x \in A, f(x) = b\}$ is a function if and only if f is one-to-one.

Exercise 2. Prove proposition 1.

Definition 3. Let $f: A \rightarrow B$ be a one-to-one function. We call the function guaranteed to exist by proposition 1, $f^{-1}: B \rightarrow A$, the *inverse* of f .

Definition 4. A function $f: A \rightarrow B$ is *onto* if for every $b \in B$ there is an $a \in A$ such that $f(a) = b$.

Example 9. $f: \mathbb{R} \rightarrow \mathbb{R}$ where $f(x) = x^2$ is onto.

Proof. If $b \in \mathbb{R}$ then $\sqrt{b} \in \mathbb{R}$, and $f(\sqrt{b}) = (\sqrt{b})^2 = b$. $\square$

Example 10. $f: \mathbb{Z} \rightarrow \mathbb{R}$ where $f(x) = x^2$ is not onto.

Proof. $2 \in \mathbb{R}$ but $\sqrt{2} \notin \mathbb{Z}$. $\square$

Exercise 3. Prove that the floor function (see example 3) is onto but not one-to-one.

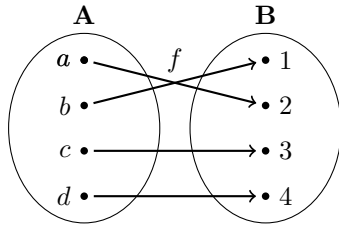


Figure 1.2:

$f: A \rightarrow B$ is one-to-one and onto.

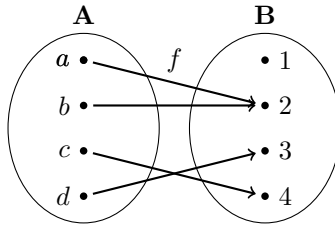


Figure 1.3:

$f: A \rightarrow B$ is not one-to-one and is not onto.

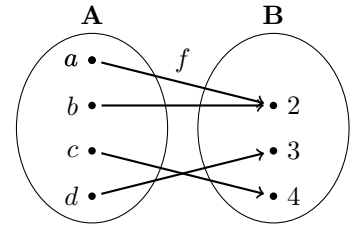


Figure 1.4:

$f: A \rightarrow B$ is onto but not one-to-one.

Definition 5. A function that is both one-to-one and onto is called a *bijection*.

Example 11. The function $f: \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = x + 1$ is a bijection.

Proof. Let $b \in \mathbb{Z}$ then $f(b-1) = b-1+1 = b$ so f is onto. Now suppose that $f(a_1) = f(a_2)$. $f(a_1) = a_1+1 = a_2+1 = f(a_2)$ implies that $a_1 = a_2$. So then f is one-to-one. $\square$

Theorem 1. Pigeonhole Principle. If $n > m$ and n items are placed in m boxes then at least one box will have two items.

Equivalently: If $f: A \rightarrow B$ and $|A| > |B|$ then f is not one-to-one.

- The pigeonhole principle says that there is at least one element $b \in B$ such that $f(a) = f(a') = b$ for some $a, a' \in A$.

Proof. Suppose to the contrary that $|A| = n > m = |B|$ and there is a one-to-one function f from A to B . If $A = \{a_1, a_2, \dots, a_n\}$ then $f(A) = \{f(a_1), f(a_2), \dots, f(a_n)\}$ lists n distinct elements since by definition $f(a_j) \neq f(a_k)$. Then $\{f(a_1), f(a_2), \dots, f(a_n)\} = \{b_1, b_2, \dots, b_n\}$ lists n distinct elements in B , but this would mean $n \leq m$ which is a contradiction. $\square$

Theorem 2. Let f be a function from A to B . The the following hold:

1. If f is one-to-one then $|B| \geq |A|$.
2. If f is onto then $|A| \geq |B|$.

3. If f is a bijection then $|A| = |B|$.

Part 1 is true by the pigeonhole principle. Part 2 and 3 are left as exercises.

Example 12. Prove parts 2 and 3 of theorem 2.

Exercise 4. Let $f: A \rightarrow B$. Prove that if f is onto then $|A| \geq |B|$.

Exercise 5. Let $f: A \rightarrow B$. Prove that if f is onto then $|A| \geq |B|$.

Exercise 6. ★ Let the function $f: A \rightarrow B$ be 1-1 and onto, i.e., f^{-1} exists and $|A| = |B|$. Prove that $(f^{-1} \circ f): A \rightarrow A$ is the identity function on A and that $(f \circ f^{-1}): B \rightarrow B$ is the identity function on B .

Exercise 7. ★ Let $f: A \rightarrow B$ be 1-1 and onto. Prove that $g = f^{-1}$, if $(g \circ f): A \rightarrow A$ is the identity function on A and $(f \circ g): B \rightarrow B$ is the identity function on B .

Exercise 8 ($\odot$). Two sets are said to be *equal* if there exists a bijective function between them; we write $|A| = |B|$. B is said to have *strictly greater cardinality* than A if there is a one-to-one function from A into B , but there does not exist a bijective function from A into B ; we write $|A| < |B|$. Prove that $|\mathbb{N}| < |\mathbb{R}|$.

1.2 Relations

In a relation, things are either related or they are not. For example, the statement “the woman x is taller than the man y ” is either true or false for any x from the set of woman and y for the set of men. This statement is either true or false. Relations are a generalization of the arithmetic relations such as $=$, $>$, $\leq$, etc.

It is different than the concept of function as it does not determine particular pairings of elements. In the example there may be many woman taller than any given man. We will investigate binary relations, meaning we will apply statements to two elements at a time.

Definition 6. Let A and B be sets and $P(a, b)$ be true or false for any $a \in A$ and $b \in B$. Then $\mathcal{R}(A, B, P(a, b))$ is called a *relation*. $P(a, b)$ is called a *statement*.

- If $P(a, b)$ is true we write $a\mathcal{R}b$ and say that a is related to b .
- If $P(a, b)$ is false we write $a\not\mathcal{R}b$ and say that a is not related to b .
- When A , B , and $P(a, b)$ are understood, we will write just $\mathcal{R}$.

Example 13. Let A be the set of men, B be the set of women, and $P(a, b)$ be the statement “ x is y ’s husband”

Then Mr. Obama $\mathcal{R}$ Mrs. Obama

Then Mr. Obama $\not\mathcal{R}$ Mr. Romney

Example 14. Let $V = \{1, 2, 3, 4\}$ and $U = \{a, b, c\}$ and $\mathcal{R} = \{(1, a), (1, b), (4, c)\}$.

$1\mathcal{R}a$, $1\mathcal{R}b$, and $4\mathcal{R}c$

$1\not\mathcal{R}c$, $2\not\mathcal{R}a$, and $3\not\mathcal{R}c$ (among many)

In the above example we did not even need to know the statement; $\mathcal{R}$ told us the relations. We can think of $\mathcal{R}$ as a set of pairs, but note that it is not a function from V to U . Since then it would map $1 \rightarrow a$ and $1 \rightarrow b$ -it is not well defined.

Exercise 9. Let $A = \{1, 2, 3, 4\}$, $B = \{5, 6\}$, and define $P(a, b) = \text{“}a \text{ divides } b\text{”}$ for $a \in A$ and $b \in B$. Find $\mathcal{R}(A, B, P)$.

Example 15. Let $A = \{1, 2, 3\}$ and define $P(a, b) = \text{“}a < b\text{”}$. Then $\mathcal{R} = \{(1, 2), (1, 3), (2, 3)\}$ since $1 < 2$, $1 < 3$, and $2 < 3$ are all true.

- If the two sets in binary relation $\mathcal{R}$ are the same, $\mathcal{R}(A, A, P)$, then we say that $\mathcal{R}$ is a *relation over A* .³

Definition 7. Let $\mathcal{R}$ be a relation over a set A then:

1. If $a\mathcal{R}a$ for all $a \in A$ then $\mathcal{R}$ is *reflexive*.

³A binary relation over a single set is also called an *endorelation*.

Use the following to answer the next two problems. Let $E = \{1, 2, 3\}$. Consider the following relations in E .

$$\begin{aligned}\mathcal{R}_1 &= \{(1, 1), (2, 1), (2, 2), (3, 2), (2, 3)\} & \mathcal{R}_4 &= \{(1, 1), (3, 2), (2, 3)\} \\ \mathcal{R}_2 &= \{(1, 1)\} & \mathcal{R}_5 &= E \times E \\ \mathcal{R}_3 &= \{(1, 2)\}\end{aligned}$$

Table 1.1: Use the above relations to complete

equivalently: $(a, a) \in \mathcal{R}$ for all $a \in \mathcal{R} \Rightarrow \mathcal{R}$ is *reflexive*.

2. If $a\mathcal{R}b$ implies that $b\mathcal{R}a$ for all $a, b \in A$ then $\mathcal{R}$ is *symmetric*.

equivalently: $(a, b) \in \mathcal{R} \Rightarrow (b, a) \in \mathcal{R}$ for all $a, b \in \mathcal{R} \Rightarrow \mathcal{R}$ is *symmetric*.

3. If $a\mathcal{R}c$ whenever $a\mathcal{R}b$ and $b\mathcal{R}c$ for all $a, b, c \in A$ then $\mathcal{R}$ is *transitive*.

equivalently: $(a, c) \in \mathcal{R}$ whenever $(a, b) \in \mathcal{R}$ and $(b, c) \in \mathcal{R}$ or all $a, b, c \in \mathcal{R} \Rightarrow \mathcal{R}$ is *transitive*.

Example 16. Let $\mathcal{R} = "|"$ over $\mathbb{Z}$; that is, let $\mathcal{R}$ be defined by the statement “ x divides y ” where $x, y \in \mathbb{Z}$.

1. Is $\mathcal{R}$ reflexive? Yes. Let $a \in \mathbb{Z}$ if $a|a$ then $a|a$.
2. Is $\mathcal{R}$ symmetric? No. $2|4$ but $4 \nmid 2$.
3. Is $\mathcal{R}$ transitive? Yes. Let $a, b, c \in \mathbb{Z}$ such that if $a|b$ and $b|c$. We can then write $c = bn$ and $b = am$ for some $m, n \in \mathbb{Z}$. If $b = am$ then $c = (am)n$ which implies that $a|c$.

Definition 8. An *equivalence relation* is a relation which is reflexive, symmetric, and transitive.

Example 17. Let $\mathcal{R}$ be “equals” over $\mathbb{R}$. That is $(a, b) \in \mathcal{R}$ if $a = b$. $\mathcal{R}$ is an equivalence relation.

Reflexive: $a = a \Rightarrow a = a$.

Symmetric: $a = b \Rightarrow b = a$.

Transitive: suppose that $a = b$ and $b = c$ then $a = b = c \Rightarrow a = c$.

Exercise 10. Each of the following statements, $P(x, y)$, define a relation R in the natural numbers, $\mathbb{N}$, i.e., $\mathcal{R} = (\mathbb{N}, \mathbb{N}, P(x, y))$. Show whether each of the following relations is reflexive.

1. $P(x, y) = x$ is less than or equal to y
2. $P(x, y) = x$ divides y
3. $x + y = 10$
4. x and y are relatively prime.

Exercise 11. Let $E = \{1, 2, 3\}$. Consider the following relations over E .

$$\begin{aligned}\mathcal{R}_1 &= \{(1, 1), (2, 1), (2, 2), (3, 2), (2, 3)\} & \mathcal{R}_4 &= \{(1, 1), (3, 2), (2, 3)\} \\ \mathcal{R}_2 &= \{(1, 1)\} & \mathcal{R}_5 &= E \times E \\ \mathcal{R}_3 &= \{(1, 2)\}\end{aligned}$$

Determine whether relation $\mathcal{R}_1, \mathcal{R}_2, \mathcal{R}_3, \mathcal{R}_4, \mathcal{R}_5$ is symmetric.

Exercise 12. Let $E = \{1, 2, 3\}$. Consider the following relations in E . Consider the following relations over E .

$$\begin{aligned}\mathcal{R}_6 &= \{(1, 2), (2, 2)\} & \mathcal{R}_9 &= \{(1, 1)\} \\ \mathcal{R}_7 &= \{(1, 2), (2, 3), (1, 3), (2, 1), (1, 1)\} & \mathcal{R}_{10} &= E \times E \\ \mathcal{R}_8 &= \{(1, 2)\}\end{aligned}$$

Determine whether each relation $\mathcal{R}_6, \mathcal{R}_7, \mathcal{R}_8, \mathcal{R}_9, \mathcal{R}_{10}$ is transitive.

Exercise 13. Define the following relation on the real numbers, $\mathcal{R} : x\mathcal{R}y \iff xy \geq 0$. Prove that $\mathcal{R}$ is reflexive and symmetric, but that $\mathcal{R}$ is not an equivalence relation.

Exercise 14. Define the following equivalence relation on the set of subsets of X as follows: $A\mathcal{R}B \iff A \cap B \neq \emptyset$ for $A, B \subseteq X$. Prove or disprove that $\mathcal{R}$ is an equivalence relation.

Let $\mathcal{R}$ and $\mathcal{R}'$ be symmetric relations in a set A . Prove that $\mathcal{R} \cap \mathcal{R}'$ is a symmetric relation in A . (note that $\mathcal{R}$ and $\mathcal{R}'$ are subsets of $A \times A$).

Exercise 15. ★ Prove: Let R and R' be symmetric relations in a set A ; then $R \cap R'$ is a symmetric relation in A .

Exercise 16. ★ Define $a\mathcal{R}b$ as $a = b \bmod n$ for $a, b \in \mathbb{Z}$ and $n \in \mathbb{Z}^+$.⁴ $a = b \bmod n$ is an equivalence relation.

⁴ $a = b \bmod n$ if $n|a - b$ for integers n, a, b . For example, $5 = 1 \bmod 4$ and $3 = 3 \bmod 4$. There is an infinite class of integers which meeting this criteria; this set is notated $[a] = \{a + kn : k \in \mathbb{Z}\}$. Modular arithmetic is covered in chapter ??.